

Nordheide Solar Park – Project Report 2024

1. Project Overview

The Nordheide Solar Park is a ground-mounted photovoltaic plant located near the village of Nordheide in Lower Saxony, Germany. It was commissioned on 14 March 2024 after a construction period of eleven months.

The project was led by Dr. Maria Keller. The engineering, procurement and construction contract was awarded to SonnenBau GmbH, a company based in Lüneburg.

The plant has an installed capacity of 18 megawatt peak (MWp) and covers an area of 22 hectares. It generates enough electricity to supply around 12,000 households.

2. Costs and Financing

The total investment for the solar park amounted to 14.5 million euros.

60 percent of the investment was financed through a loan from Regionalbank Lüneburg with a fixed interest rate of 3.2 percent and a term of 18 years.

The remaining 40 percent was raised through a citizen participation scheme. A total of 830 local residents invested in the project. The minimum investment was 500 euros, and investors receive an expected annual return of 4.1 percent.

3. Technology

The park consists of 41,200 bifacial solar modules of the type HX-440, manufactured by Helion Solar. The modules are mounted on single-axis trackers that follow the sun from east to west.

Six central inverters convert the direct current into alternating current.

The park is connected to the grid via the 20 kV substation in Bardowick. The underground connection cable has a length of 3.4 kilometres.

4. Operations in 2024

Between commissioning and 31 December 2024, the park generated 16.8 gigawatt hours (GWh) of electricity. The technical availability was 99.1 percent.

There were seven unplanned outages in 2024. The longest outage lasted eleven hours on 3 August 2024 and was caused by a fault in one of the inverters.

Operations and maintenance are carried out by NordService, which employs three technicians on site.

5. Environment and Biodiversity

Around 240 sheep from the local shepherd Jan Brügge graze between the module rows and keep the vegetation short.

Wildflower strips with a total length of 1.8 kilometres were planted along the fences to support insects.

A bird survey in 2024 counted 23 bird species on the site, compared to 14 species in the baseline survey of 2022. The park avoids around 9,400 tonnes of CO₂ emissions per year.

6. Outlook

A battery storage system with a capacity of 10 megawatt hours (MWh) is planned for 2026. Its estimated cost is 3.2 million euros.

A second expansion phase will add 6 MWp on an adjacent area of 7 hectares. The building permit is expected in 2027.

From 2025 onwards, the operator pays 25,000 euros per year into a community fund for the village of Nordheide.